

1. What is Exception Handling in Python?

Ans:

Exception Handling is a mechanism in Python used to handle runtime errors (exceptions) so that the program does not terminate unexpectedly. It allows the program to continue executing normally even if an error occurs.

Need for Exception Handling:

- Prevents the program from crashing.
- Handles unexpected errors gracefully.
- Improves program reliability and robustness.
- Allows developers to display meaningful error messages.

Example:

```
try:
    num = 10 / 0
except ZeroDivisionError:
    print("Cannot divide by zero.")
```

Output:

Cannot divide by zero.

2. How do you use the `try`, `except`, and `finally` blocks for handling exceptions in Python?

Ans:

Python uses the `try`, `except`, and `finally` blocks to handle exceptions.

- **`try` block:** Contains the code that may cause an exception.
- **`except` block:** Executes if an exception occurs in the `try` block.
- **`finally` block:** Executes regardless of whether an exception occurs or not. It is commonly used to close files or release resources.

Syntax:

```
try:
    # Code that may raise an exception
except ExceptionType:
    # Code to handle the exception
finally:
    # Code that always executes
```

Example:

```
try:
    num1 = 10
    num2 = 0
    result = num1 / num2
    print(result)

except ZeroDivisionError:
    print("Error: Division by zero is not allowed.")

finally:
    print("Program execution completed.")
```

Output:

Error: Division by zero is not allowed.
Program execution completed.